Yug Pathak

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SUMMARY

Computer Science undergraduate with a **8.88/10 CGPA** and hands-on experience in **software engineering, backend development, full-stack development, REST APIs, AI/LLM systems, databases, and enterprise applications**. Proficient in **Python, C++, Java, JavaScript, TypeScript, FastAPI, Flask, React, Next.js, Node.js, SQL, PostgreSQL, SQLAlchemy, Docker, Git, and cloud technologies**. Strong foundation in **Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, and Software Engineering**.

TECHNICAL SKILLS

Programming Languages: C++, Python, Java, JavaScript, TypeScript, SQL

Backend Development: FastAPI, Flask, Django, Node.js, Express.js, REST APIs, Pydantic, SQLAlchemy, Alembic, WebSockets

Frontend Development: React, Next.js, TypeScript, HTML5, CSS3, Tailwind CSS, Zustand

Databases: PostgreSQL, SAP HANA, Firebase, Supabase, SQLAlchemy, Prisma

AI and Machine Learning: Large Language Models, LLM Evaluation, RAG, RLHF, SFT, AI Data Annotation, Human Feedback, Prompt Evaluation

Cloud and DevOps: Docker, Git, GitHub, Vercel, Render, Redis, Celery, RabbitMQ

Security and Distributed Systems: JWT, OAuth2, RBAC, WebSockets, Web3, Smart Contracts, MetaMask

Core Computer Science: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering

EDUCATION

B.Tech in Computer Science and Engineering, Poornima University 08/2024 – 08/2028 | Jaipur, Rajasthan
CGPA: **8.88/10**

EXPERIENCE

Deccan AI, RLHF and SFT AI Trainer (Freelance) 11/2025 – Present | Remote

- Contributed to **5 AI annotation and LLM evaluation projects** involving response evaluation, AI data collection, human feedback, and multimodal data annotation.
- Evaluated and compared **LLM-generated responses** for factuality, relevance, user intent, instruction following, and overall response quality using structured evaluation rubrics.
- Reviewed **Python and Machine Learning code generated by LLMs**, validating correctness, implementation quality, library usage, and alignment with technical requirements.

Mittal Alliance Industries Pvt. Ltd., Frontend Developer Intern 03/2026 – Present | Remote

- Delivered **production-grade software solutions** for **3 international clients and 1 domestic client**.
- Developed and integrated **10+ REST APIs** supporting **500+ daily interactions** across production applications.
- Built **graph-based route optimization systems**, reducing travel distance by **22%** and operational costs by **18%**.

PROJECTS

Resilience OS, AI-Powered Supply Chain Disruption Control and Simulation Platform resilience-os.vercel.app/command-center

- Architected and developed an **AI-powered supply chain disruption control and simulation platform** using **Next.js, React, Zustand, FastAPI, Python, and SQLAlchemy** to simulate disruptions and evaluate recovery strategies.
- Integrated an enterprise-grade **SAP Cloud Application Programming Model (CAP) Node.js backend** with **SAP HANA** using **sqlalchemy-hana** for secure and scalable supply chain data management.
- Developed an interactive **Command Center dashboard** with **@xyflow/react** workflow visualization, **Recharts** analytics, and automated PDF report generation using **jsPDF**.
- Designed a **hallucination-protected AI Adapter** connecting deterministic simulation engines with **Large Language Models (LLMs)**, restricting recommendations to pre-calculated feasible disruption recovery options.

Oracle Before Signing, Blockchain Transaction Security System obs-dev.vercel.app

- Built a **permission-first transaction validation system** with **5+ smart contracts** on the **Monad blockchain**, reducing unsafe transactions by approximately **90%**.
- Implemented **MetaMask authentication, Web3 transaction workflows, authorization controls, and adversarial validation** to mitigate unauthorized blockchain transactions.

Campus OS, AI-Powered Multi-Tenant Campus Management Platform campus-os-beta.vercel.app

- Built an **AI-powered full-stack campus management platform** using **Next.js, React, TypeScript, Node.js, Express.js, and PostgreSQL**.
- Developed **REST APIs, JWT-based RBAC, Prisma database architecture, Redis integration, and Docker deployment**, with **LLM and RAG** for intelligent document search and campus assistance.

ACHIEVEMENTS

- **Oracle Cloud Infrastructure Certified Developer Professional 2025**.
- **Finalist, SAP HackFest 2026 National Hackathon**.
- **Semifinalist, SCPC–Henkel National Supply Chain Hackathon**; advanced to finals at **SP Jain Institute of Management and Research, Mumbai**.
- **Top 6, Reskill SprintForGood AI Hackathon, IIT Delhi**.
- **Top 5, LNMHacks 8.0 Blockchain Hackathon**, among 100+ teams.
- Advanced to the **Online Assessment rounds of Amazon ML Summer School and Flipkart GRID 8.0**.

LEADERSHIP & INVOLVEMENT

Google Developer Groups On Campus, Organiser/Lead 08/2025 – Present | Jaipur, Rajasthan

- Selected as the official **Google Developer Groups On Campus Organiser of Poornima University** from a competitive pool of applicants.